

CSE 4631

Digital Signal Processing

Week 12

Linearity of the Fourier Transform

The Fourier Transform is linear, that is, it possesses the properties of *homogeneity* and *additivity*. This is true for all four members of the Fourier Transform family.

Homogeneity

A change in amplitude in one domain produces an identical change in the amplitude in the other domain.

$$\text{If: } x[t] \longrightarrow X[f]$$

$$\text{Then: } kx[t] \longrightarrow kX[f]$$

In rectangular notation $kX[f]$ means both the real part and imaginary part are multiplied by k

In polar notation it means that the magnitude is multiplied by k , while the phase remains unchanged

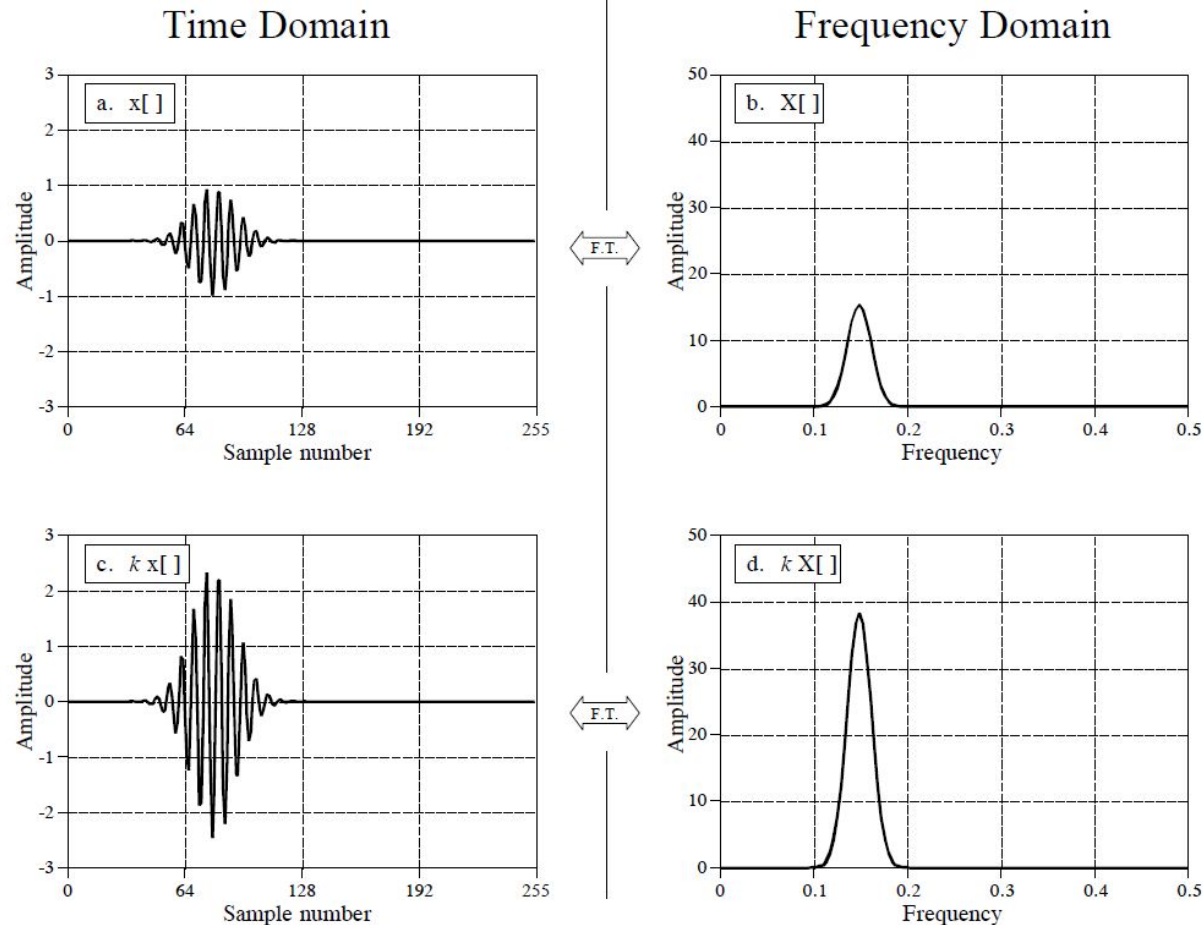


FIGURE 10-1

Homogeneity of the Fourier transform. If the amplitude is changed in one domain, it is changed by the same amount in the other domain. In other words, *scaling* in one domain corresponds to *scaling* in the other domain.

Linearity of the Fourier Transform (contd.)

Additivity

Addition in one domain corresponds to addition in the other domain.

$$\text{If: } x_1[n] + x_2[n] = x_3[n]$$

$$\text{Then: } \text{Re}X_1[f] + \text{Re}X_2[f] = \text{Re}X_3[f] \text{ and } \text{Im}X_1[f] + \text{Im}X_2[f] = \text{Im}X_3[f]$$

Frequency spectra in polar form cannot be directly added; they must be converted into rectangular notation, added, then reconverted back to polar form.

Suppose you are adding two sinusoids having the same frequency, but with different amplitudes and phases.

- If the two phases are same, the amplitudes will add
- If the two phases are exactly opposite, the amplitudes will be subtracted

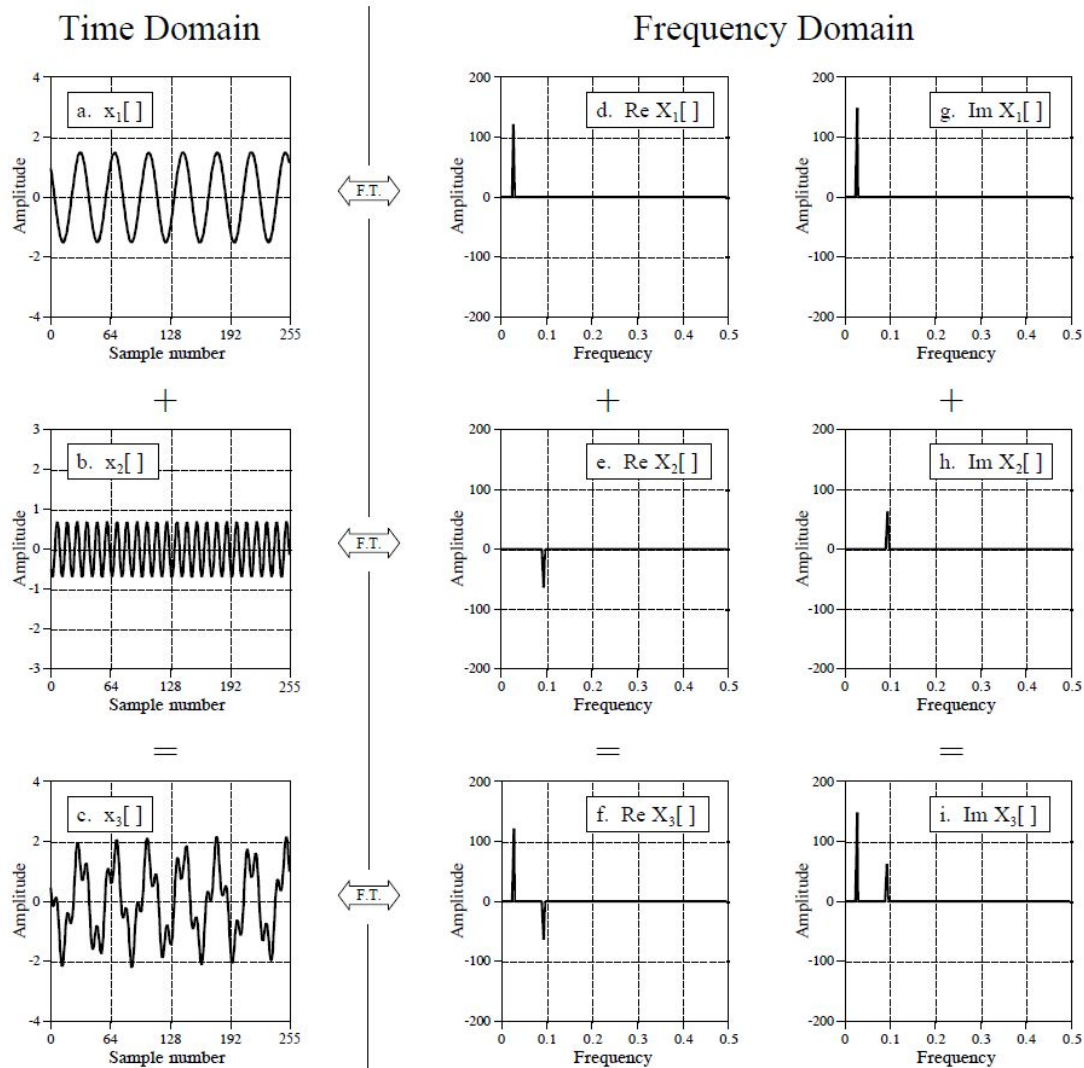
Hence, you cannot simply add the magnitudes and phases when in polar form

In spite of being linear, **the Fourier transform is not shift invariant**. In other words, a shift in the time domain does not correspond to a shift in the frequency domain.

Then how does the frequency domain react to a shift in time domain?

Linearity of the Fourier Transform (contd.)

Additivity



Characteristics of the Phase

If: $x[n] \Leftrightarrow \text{Mag}X[f] \ \& \ \text{Phase}X[f]$

Then: $x[n+s] \Leftrightarrow \text{Mag}X[f] \ \& \ \text{Phase}X[f] + 2\pi sf$

Here f is expressed as a fraction of the sampling rate, running between 0 and 0.5

Characteristics of the Phase (contd.)

- The time domain waveform is symmetrical around a vertical axis, that is, the left and right sides are mirror images of each other – **phase of their frequency spectrum is a straight line – linear phase.**
- When the time domain signal is shifted to the right, there is a decrease in the slope (**c, g**)
- When the time domain signal is shifted to the left, there is an increase in the slope (**a, e**)
- If the time domain signal is symmetrical around sample zero, the phase is entirely zero.
- Since, DFT views the time domain as circular with sample zero inherently connected to sample $N-1$ any signal symmetrical around sample zero will also be symmetrical around sample $N/2$, and vice versa. (**b, f**)

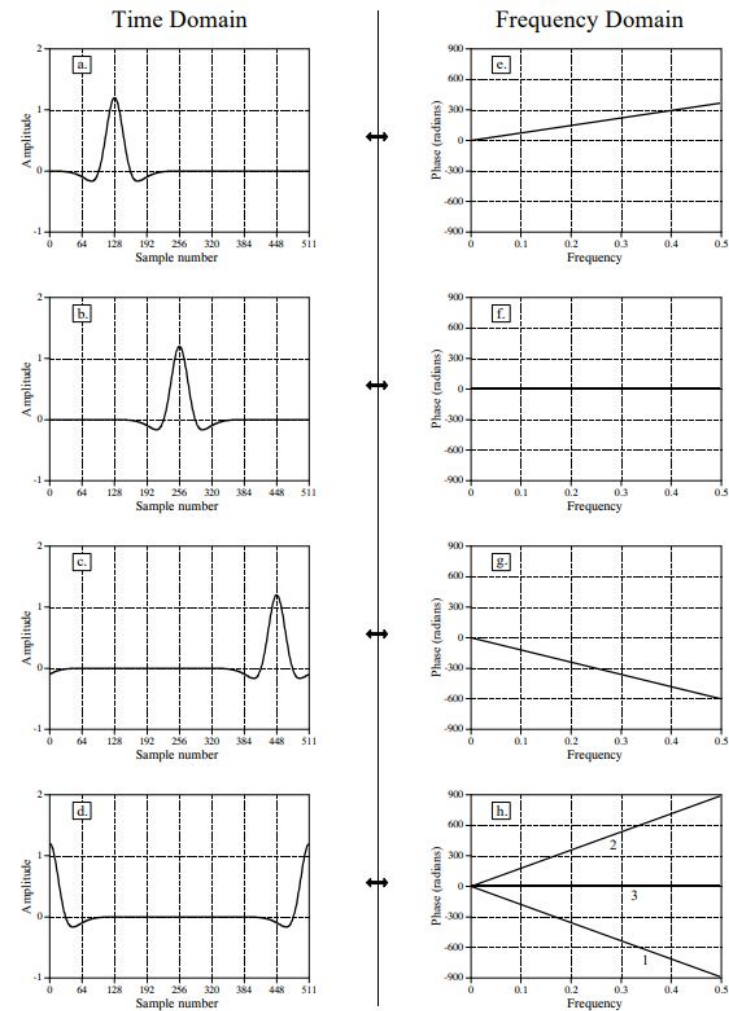


FIGURE 10-3
Phase changes resulting from a time domain shift.

Characteristics of the Phase (contd.)

- In (d, h), which of these three phases are correct? **They all are** depending on how the π and 2π phase ambiguities are arranged.
- For instance, every sample in line 2 differs from the corresponding sample in line 1 by an integer multiple of 2π , making them equal.
- To relate line 3 to lines 1 and 2, the π ambiguities must also be taken into account.

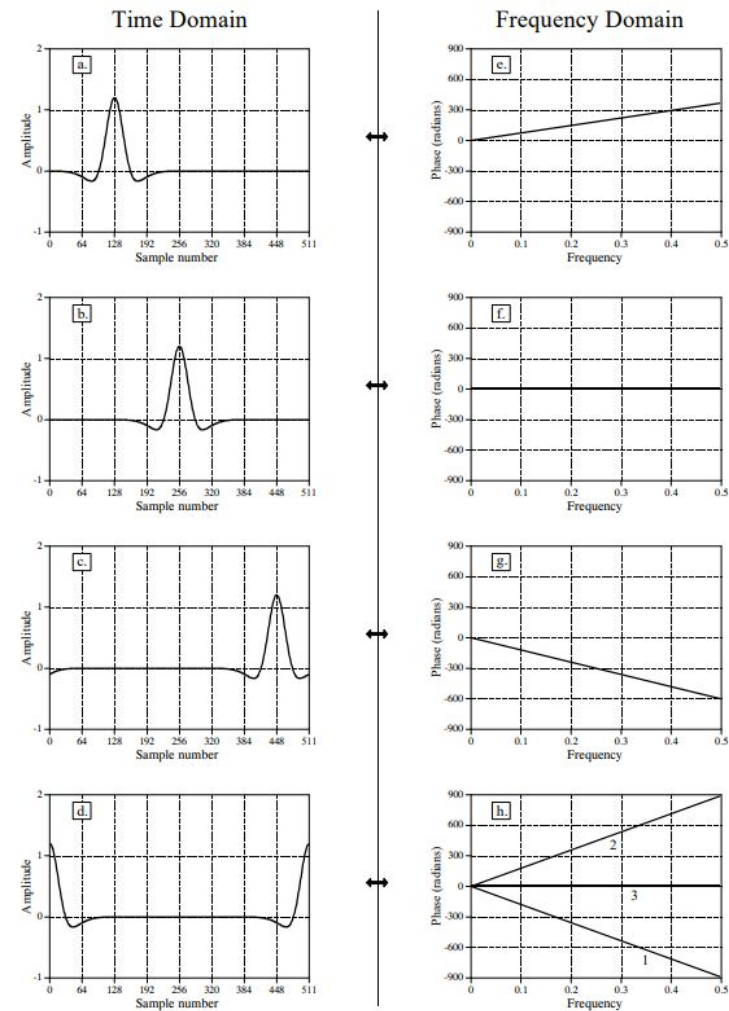


FIGURE 10-3
Phase changes resulting from a time domain shift.

Characteristics of the Phase (contd.)

A shift, when expressed in terms of a phase change, is proportional to the frequency of the sinusoid Being shifted ($-2\pi fn$ radians phase shift; f = frequency; n = number of samples getting shifted)

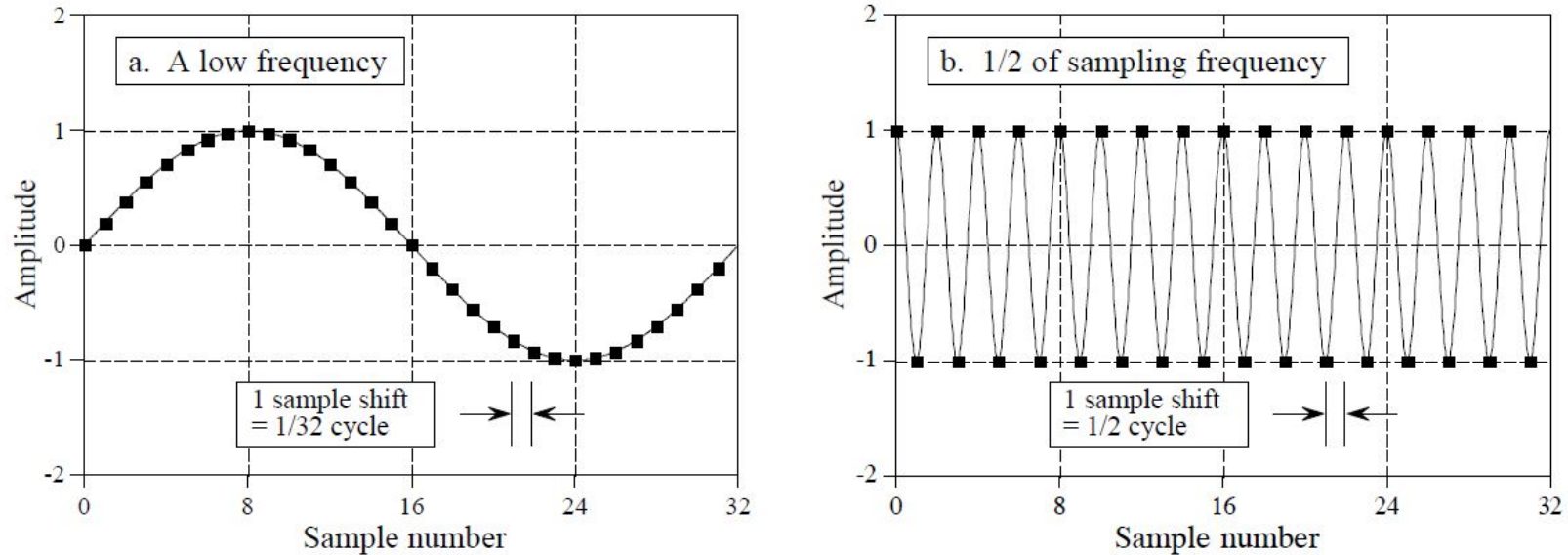


FIGURE 10-4

The relationship between samples and phase. Figures (a) and (b) show low and high frequency sinusoids, respectively. In (a), a one sample shift is equal to 1/32 of a cycle. In (b), a one sample shift is equal to 1/2 of a cycle. This is why a shift in the waveform changes the phase more at high frequencies than at low frequencies.

Characteristics of the Phase (contd.)

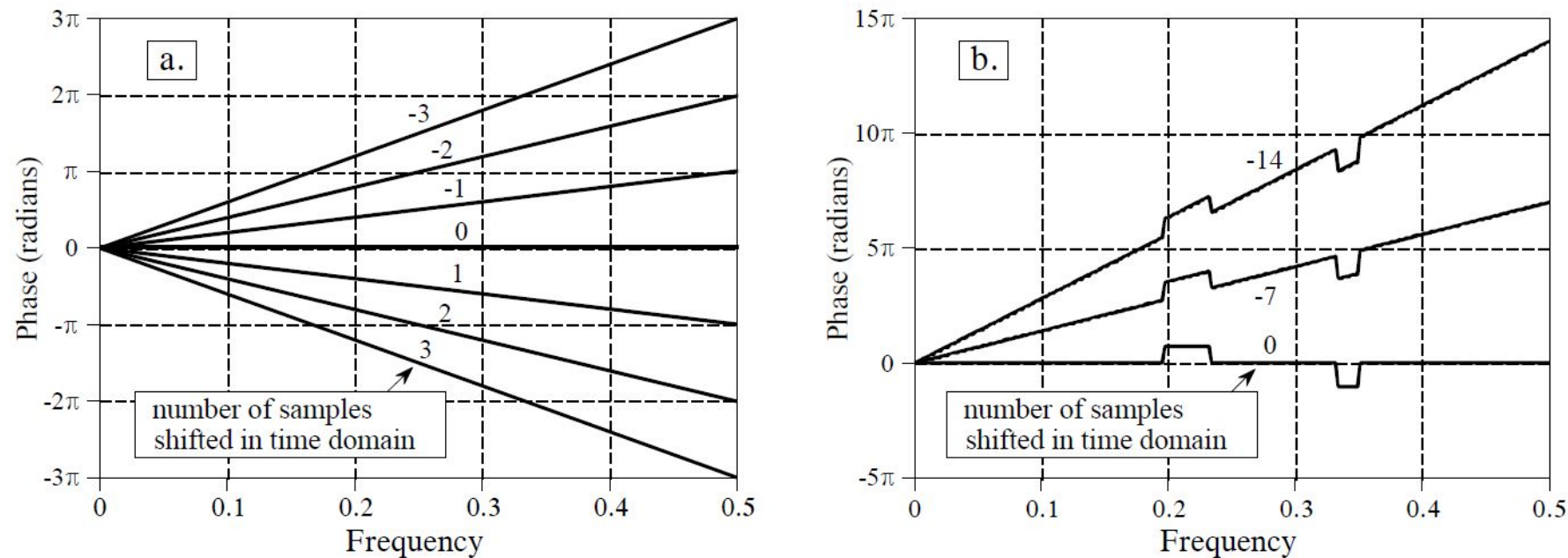


FIGURE 10-5
Phases resulting from time domain shifting. For each sample that a time domain signal is shifted in the positive direction (i.e., to the right), the phase at frequency 0.5 will decrease by π radians. For each sample shifted in the negative direction (i.e., to the left), the phase at frequency 0.5 will increase by π radians. Figure (a) shows this for a linear phase (a straight line), while (b) is an example using a nonlinear phase.

Characteristics of the Phase (contd.)

Question: What information is contained in the phase, and what information is contained in the magnitude?

- The phase is replaced with random numbers between $-\pi$ and π before applying inverse DFT to reconstruct the time domain waveform in **(b)**
- The magnitude is replaced with small random numbers before applying inverse DFT in **(c)**
- **(b)** is based only on the information contained in the *magnitude* and **(c)** is based only on the information contained in the *phase*

Much of the information about the shape of the time domain waveform is contained in the phase, rather than the magnitude.

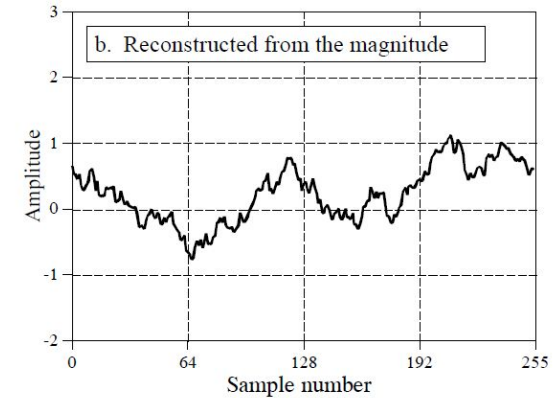
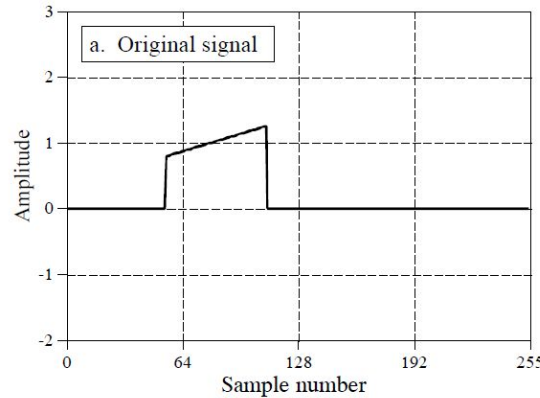
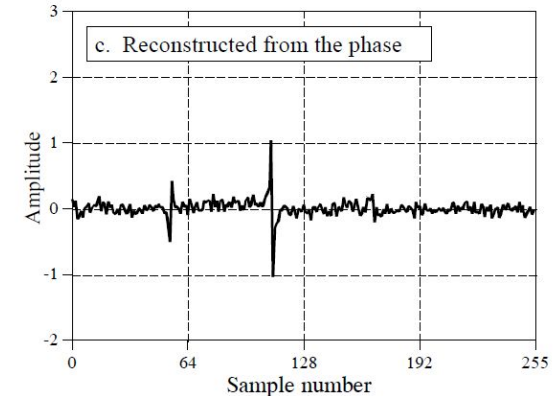


FIGURE 10-6 Information contained in the phase. Figure (a) shows a pulse-like waveform. The signal in (b) is created by taking the DFT of (a), replacing the *phase* with random numbers, and taking the Inverse DFT. The signal in (c) is found by taking the DFT of (a), replacing the *magnitude* with random numbers, and taking the Inverse DFT. The location of the *edges* is retained in (c), but not in (b). This shows that the phase contains information on the location of events in the time domain signal.



Characteristics of the Phase (contd.)

Question: Why does left-right symmetry correspond to a zero (or linear) phase?

- Decompose the original time domain signal into two halves which are perfect mirror images of each other
- In frequency domain, their magnitudes will be identical, while the phases will be opposite in sign

Two important concepts from this:

1. The nonlinear phase of the left half exactly cancels out the nonlinear phase of the right half
2. If you flip the left side so that it becomes identical to the right side, the magnitude remains the same, but it changes the sign of every point in the phase, and vice versa.

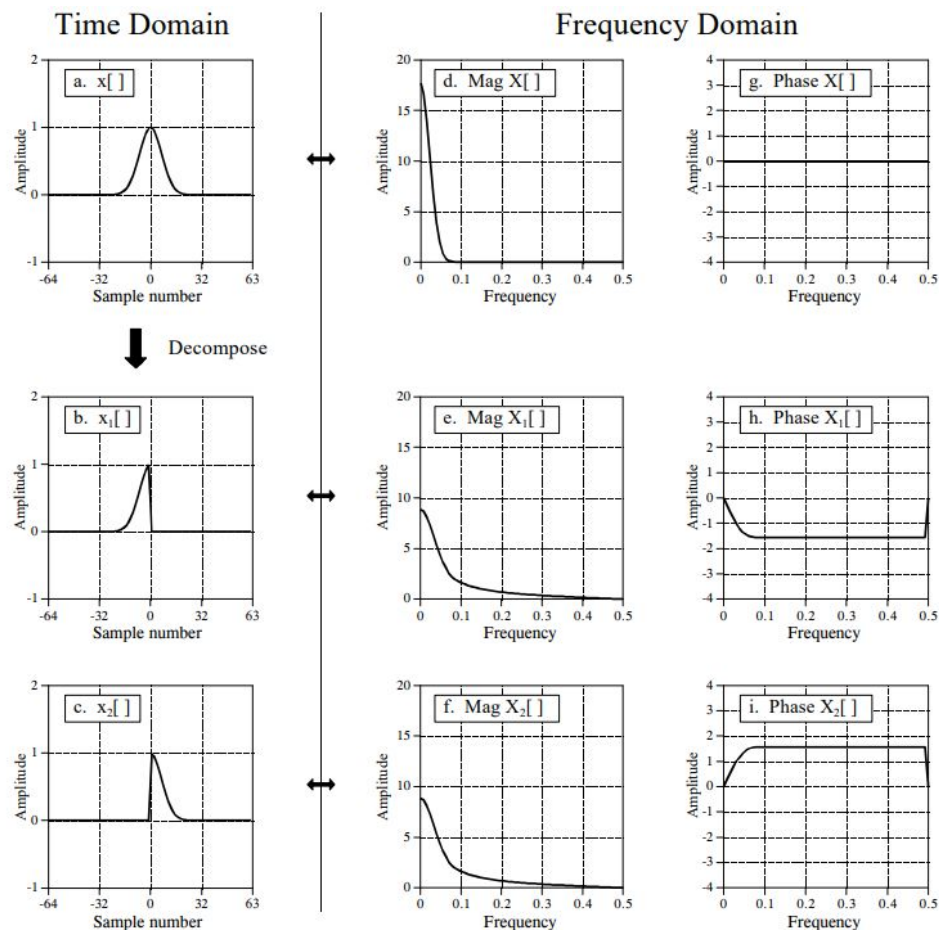


FIGURE 10-7

Phase characteristics of left-right symmetry. A signal with left-right symmetry, shown in (a), can be decomposed into a right half, (b), and a left half, (c). The magnitudes of the two halves are identical, (e) and (f), while the phases are the negative of each other, (h) and (i).

Characteristics of the Phase (contd.)

Complex Conjugation

The operation of changing the sign of the phase in frequency domain which flips the time domain signal left-for-right.

In mathematical terms (polar form):

If $X[f]$ consists of $Mag X[f]$ and $Phase X[f]$, then $X^*[f]$ is called the *complex conjugate* and is composed of $MagX[f]$ and $-PhaseX[f]$

In mathematical terms (rectangular form):

If $X[f]$ is composed of $ReX[f]$ and $ImX[f]$, then the *complex conjugate* $X^*[f]$ is made up of $ReX[f]$ and $-ImX[f]$

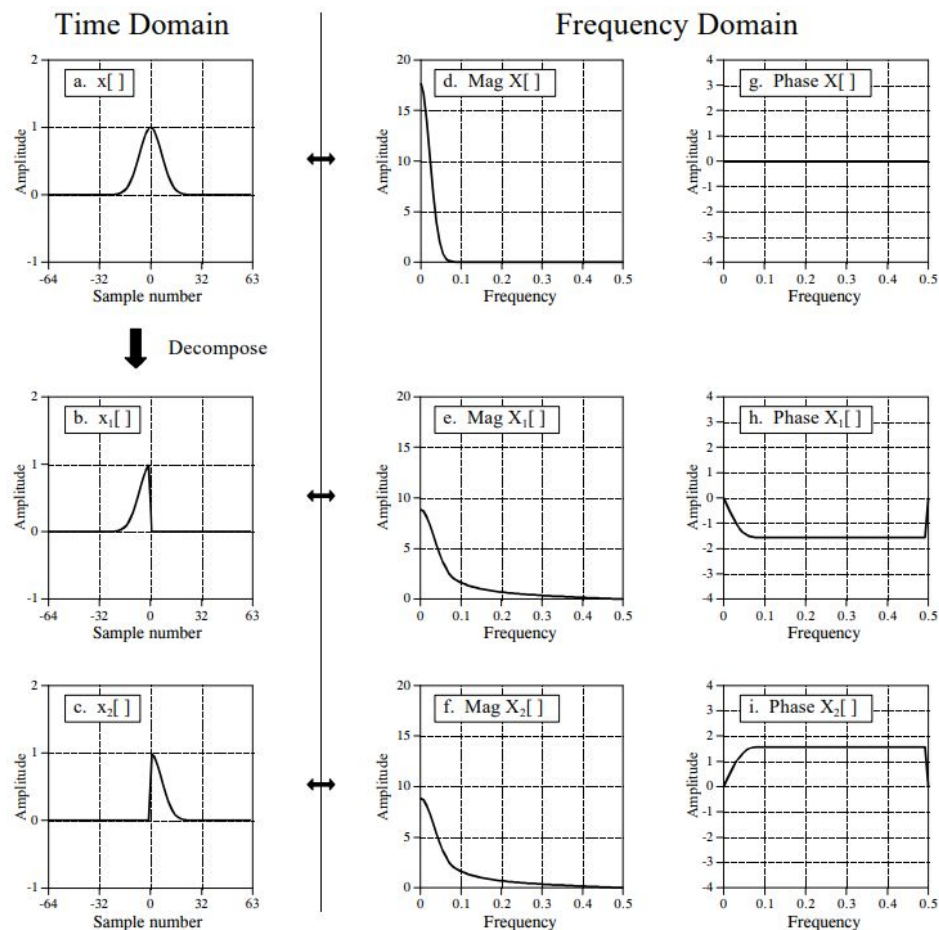


FIGURE 10-7

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Characteristics of the Phase (contd.)

How can we use complex conjugate in DSP?

- If $x[n]$ has a Fourier transform of $X[f]$, then $x[-n]$ has a Fourier transform of $X^*[f]$
- Correlation and convolution:
 - In time domain, $a[n] * b[n]$ is convolution while $a[n] * b[-n]$ is correlation
 - In frequency domain they correspond to $A[f] \times B[f]$ and $A[f] \times B^*[f]$
- The frequency spectrum of a signal can be changed to zero phase by multiplying it by its complex conjugate: $X[f] \times X^*[f]$
 - Whatever phase happens to have will be canceled by adding its opposite (when frequency spectra are multiplied, the phases are added)
- In time domain, $x[n] * x[-n]$ will have a left-right symmetry around sample zero, regardless of what $x[n]$ is.

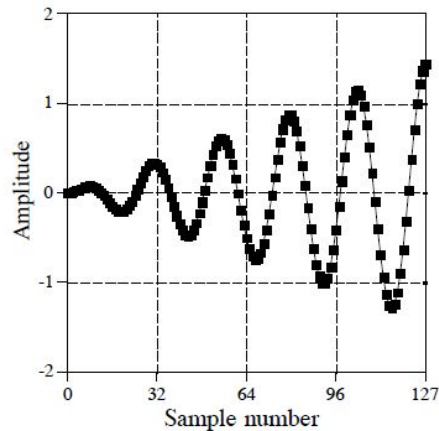
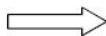
Periodic Nature of the DFT

Periodicity in the time domain

- Unlike the other three Fourier Transforms, the DFT views both the time domain and the frequency domain as periodic.
- If you have an N point signal in time domain, DFT views those N points to be a single period of an infinitely long periodic signal. This means that –
 - The left side of the signal is connected to the right side of a duplicate signal.
 - Likewise, the right side of the signal is connected to the left side of an identical signal.
- In this view, sample $N-1$ occurs next to sample 0. This is referred to as being **circular**.
- The most serious consequence of time domain periodicity is **time domain aliasing**.

Periodic Nature of the DFT (contd.)

The time domain
viewed as N points



The time domain
viewed as periodic

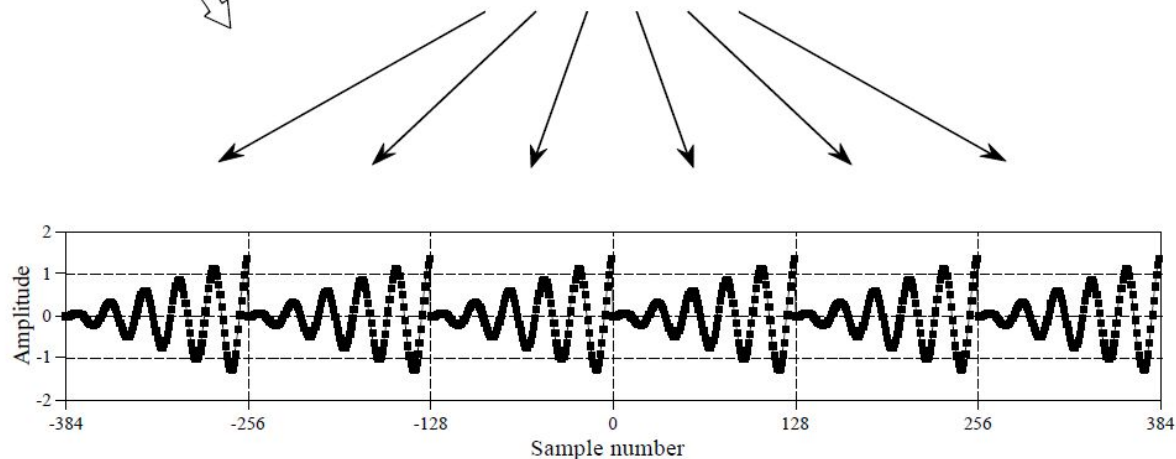


FIGURE 10-8

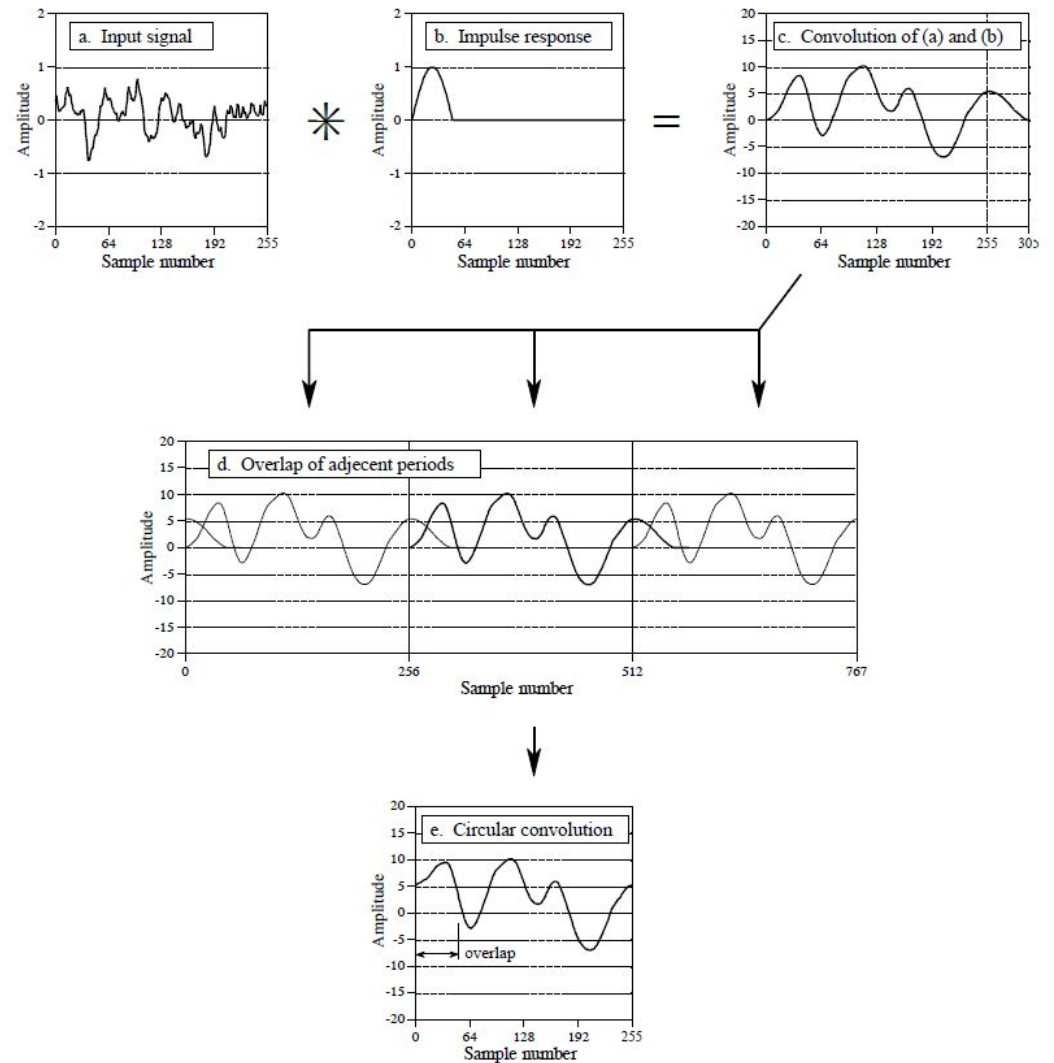
Periodicity of the DFT's time domain signal. The time domain can be viewed as N samples in length, shown in the upper figure, or as an infinitely long periodic signal, shown in the lower figure.

Periodic Nature of the DFT (contd.)

Time domain aliasing

- Suppose you have a time domain signal and you have passed it through the DFT to find its frequency spectrum.
- Then you modify the frequency spectrum in some manner before using the inverse DFT.
- Some of the changes routinely done in DSP include: deleting selected frequencies, changing the amplitude, shifting around, etc.
- These changes can create a time domain signal that is too long to fit into a single period which forces the signal to spill over from one period to the adjacent periods.
- When time domain is viewed as circular, portions of the signals that overflow on the right suddenly seems to reappear on the left side of the signal, and vice versa.
- This means that the overflowing signal **alias** themselves to a new location in the time domain.
- If this new location already contain an existing signal, the whole mess adds, resulting in a loss of information.
- Example: circular convolution resulting from frequency domain multiplication!

Periodic Nature of the DFT (contd.)



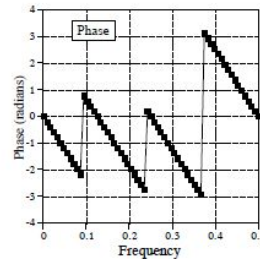
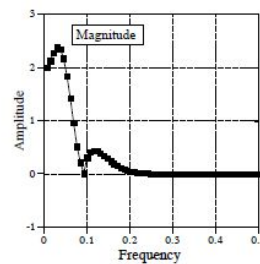
Periodic Nature of the DFT (contd.)

Periodicity in the frequency domain

- The key feature in the periodicity in the frequency domain is that the frequency spectrum between 0 and 0.5 appears to have a mirror image of frequencies that run between 0 and -0.5.
- This mirror image of negative frequencies is *slightly different* for the magnitude and the phase signals.
- In the magnitude, the signal is flipped left-for-right.
- In the phase, the signal is flipped left-for-right, *and* changes in sign.
- Therefore, the magnitude is said to be an even signal (with even symmetry)
- And the phase is said to be an odd signal (with odd symmetry)
- Taking these negative frequencies into account, the DFT views the frequency domain as periodic, with a period of 1.0 times the sampling rate, such as -0.5 to 0.5, or 0 to 1.0.
- In terms of sample numbers, this makes the length of the frequency domain period equal to N , the same as in the time domain.

Periodic Nature of the DFT (contd.)

The frequency domain viewed as 0 to 0.5 of the sampling rate



The frequency domain viewed as periodic

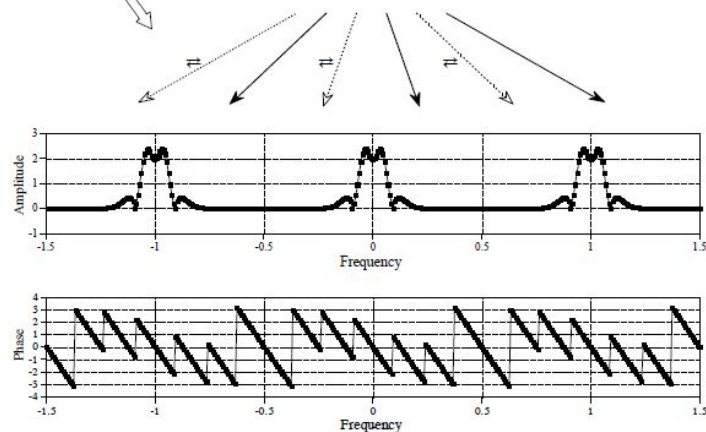


FIGURE 10-9

Periodicity of the DFT's frequency domain. The frequency domain can be viewed as running from 0 to 0.5 of the sampling rate (upper two figures), or an infinity long periodic signal with every other 0 to 0.5 segment flipped left-for-right (lower two figures).

Periodic Nature of the DFT (contd.)

Frequency domain aliasing

- The periodicity of the frequency domain makes it susceptible to frequency domain aliasing.
- Imagine a time domain signal that corresponds to some frequency spectrum. If the time domain signal is modified, it is obvious that the frequency spectrum will also be changed.
- If the modified frequency spectrum cannot fit in the space provided, it will push into the adjacent periods.
- Just as before, this aliasing causes two problems: frequencies aren't where they should be, and overlapping frequencies from different periods add, destroying information.
- Frequency domain aliasing is more difficult to understand than time domain aliasing due to the complicated periodic pattern in the frequency domain.

Periodic Nature of the DFT (contd.)

Frequency domain aliasing (contd.)

- Consider a single frequency that is being forced to move from 0.01 to 0.49 in the frequency domain. The corresponding negative frequency is therefore moving from -0.01 to -0.49. (mirroring)
- In this way, if the positive frequency moves across the 0.5 barrier, the negative frequency is pushed across the -0.5 barrier.
- Since the frequency domain is periodic, these same events are occurring in the other periods, such as between 0.5 and 1.5.
- A clone of the positive frequency is crossing frequency 1.5 from left to right, while a clone of the negative frequency is crossing 0.5 from right to left.
- If you can only see the frequency band of 0 to 0.5, it appears that the frequency leaving to the right, reappears on the right, but moving in the opposite direction.

Periodic Nature of the DFT (contd.)

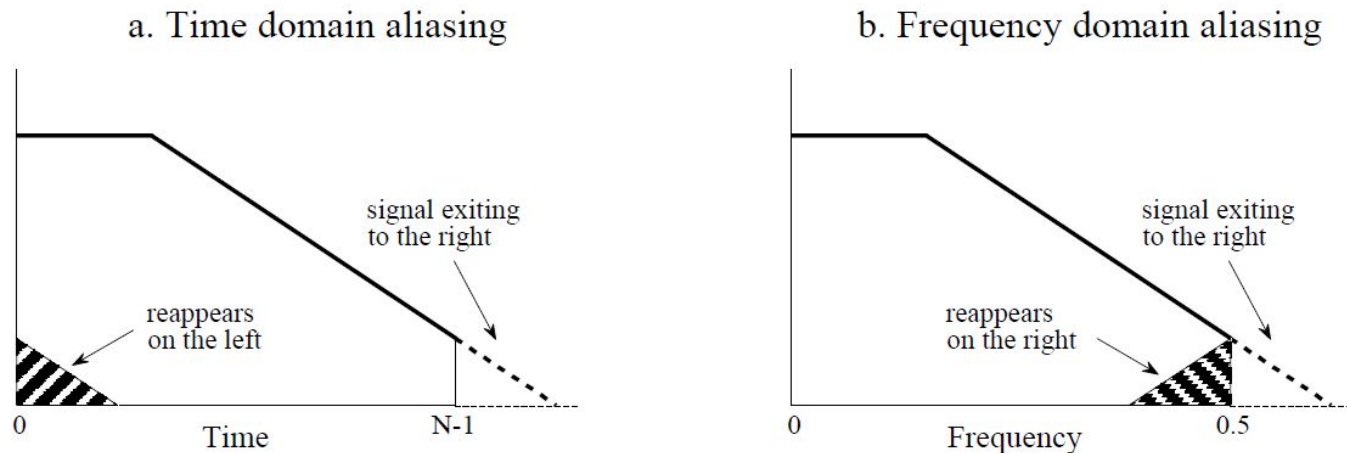


FIGURE 10-10

Examples of aliasing in the time and frequency domains, when only a single period is considered. In the time domain, shown in (a), portions of the signal that exit to the right, reappear on the left. In the frequency domain, (b), portions of the signal that exit to the right, reappear on the right as if they had been folded over.

Periodic Nature of the DFT (contd.)

Negative frequencies (explanation of the figure in the next slide)

- Suppose you are given a discrete signal composed of 32 samples that represent a discrete cosine wave. Therefore, $x(n) = \cos(2\pi n f / N + \theta)$
- You are given the task of finding the frequency spectrum that corresponds to these 32 points.
- After a while, you figured that $f=3$ and $\theta = -\pi/4$. However, there are other solutions as well!
- Firstly, $f=-3$ and $\theta = \pi/4$, which is a mathematically valid solution. That means, every *positive* frequency sinusoid can alternatively be expressed as a *negative* frequency sinusoid.
- Secondly, $f=35$ and $\theta = -\pi/4$ passes through all of the discrete points, and is therefore a correct solution as well. Likewise, $f=\pm 29$, $f=\pm 35$, $f=\pm 61$, and $f=\pm 67$ are all valid solutions.
- Note that although these solutions make the signal oscillate multiple times between the given points doesn't disqualify them from being correct.
- The first solution corresponds to frequencies between 0 and 0.5 of the sampling rate. The second one results in frequencies between 0 and -0.5. Lastly, the rest of the solutions makes up the infinite number of duplicate frequencies below -0.5 and above 0.5

Periodic Nature of the DFT (contd.)

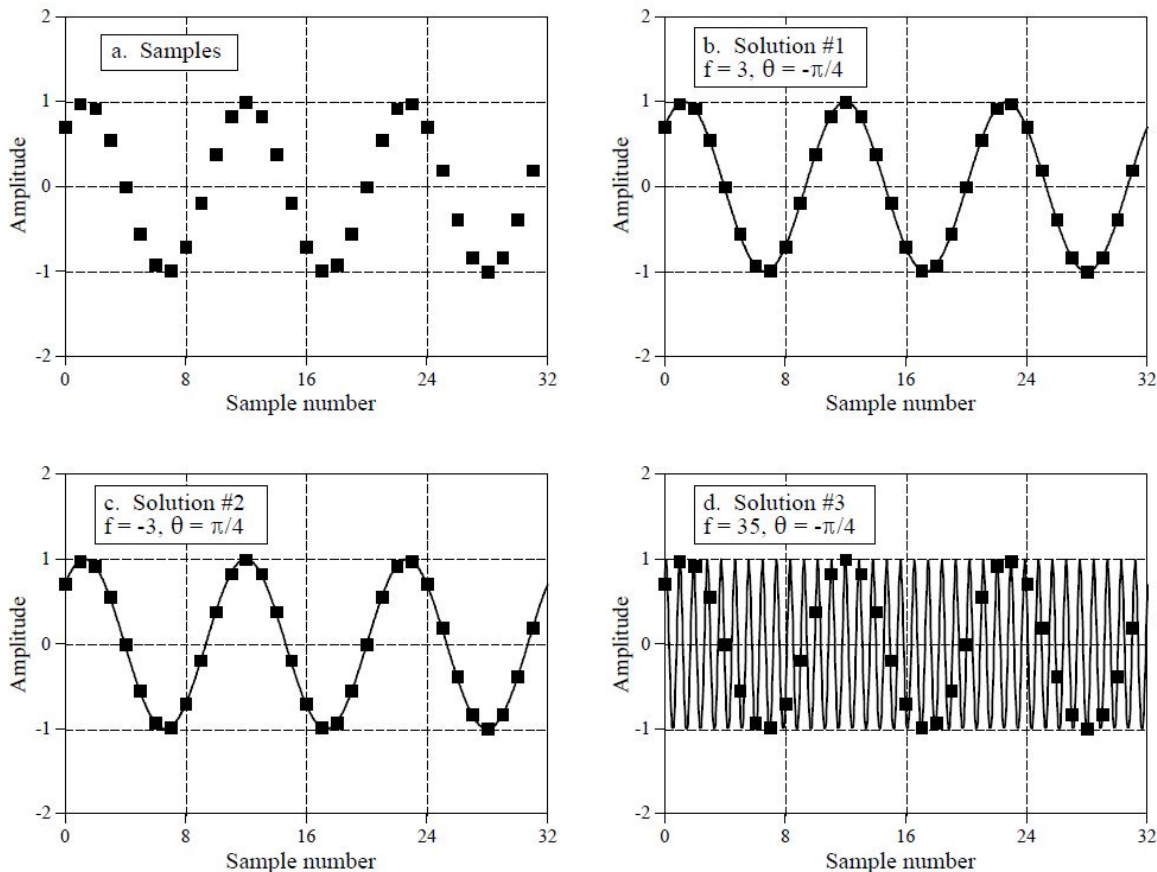


FIGURE 10-11

The meaning of negative frequencies. The problem is to find the frequency spectrum of the discrete signal shown in (a). That is, we want to find the frequency and phase of the sinusoid that passed through all of the samples. Figure (b) is a solution using a *positive* frequency, while (c) is a solution using a *negative* frequency. Figure (d) represents a family of solutions to the problem.

Periodic Nature of the DFT (contd.)

Use of aliasing or negative frequencies

- Many DSP techniques do not require the use of negative frequencies, or an understanding of the DFT's periodicity. For these applications, it is sufficient to view the time domain as extending from 0 to $N-1$, and the frequency domain as from 0 to 0.5 of the sampling frequency.. Such as,
 - Spectral analysis
 - Frequency response of systems
- However, certain procedures can only be analyzed by considering how signals overflow between periods. Such as,
 - Circular convolution: time domain aliasing
 - Analog-to-digital conversion: frequency domain aliasing